Comprehensive Attribute Difference Attention Network for Remote Sensing Image Semantic Understanding

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***Abstract*— In the task of semantic understanding of remote sensing images, most current research focuses on learning contextual information through attention mechanisms or multiple inductive biases. However, these methods are limited in capturing fine-grained differences within the same attribute, are susceptible to background noise interference, and lack effective modeling capabilities for spatial relationships and long-range dependencies between different remote sensing attributes. To address these issues, we specifically focus on the homogeneous and heterogeneous differences between attributes in remote sensing images. Thus, we propose an innovative comprehensive attribute difference attention network (CADANet) to enhance the performance of understanding remote sensing images. Specifically, we design two key modules: the attribute feature aggregation (AFA) module and the context attribute-aware spatial attention (CAASA) module. The AFA module primarily focuses on global and local domain attribute modeling, reducing the impact of homogeneous attribute differences through fine-grained feature extraction and global context information. The CAASA module integrates pixel-level global background information and relative position priors, employing a self-attention mechanism to capture long-range dependencies, thus addressing heterogeneous attribute differences. Extensive experimental results conducted on the widely used Vaihingen, Potsdam, and WHDLD datasets effectively demonstrate that our proposed method outperforms other recent approaches in performance. Our code is available at https://github.com/lzp-lkd/CADANet.**

***Index Terms*— Attention mechanism, attribute perception, context modeling, feature extraction, multiattribute scene understanding.**

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I. Introduction

IN THE expansive field of Earth remote sensing attribute understanding, the importance of semantic segmentation of remote sensing images is increasingly prominent. This task, which involves accurately assigning each pixel to a specific semantic attribute such as roads, buildings, and water bodies, plays a pivotal role not only in critical areas, such as urban planning [1], [2], environmental monitoring [3], [4], [5], disaster response [2], [6], [7], and land-cover analysis [8], [9], [10], [11] but also in advancing the automation of surface spatial information generation. With the rapid advancement of remote sensing technology, high-resolution remote sensing images enable unprecedented detail and scope in Earth observation, presenting both unparalleled opportunities and challenges for semantic understanding of remote sensing images.

Despite the significant successes of convolutional neural networks (CNNs) in image feature extraction and processing, which have greatly propelled the development of remote sensing image analysis technology, they exhibit notable limitations when handling remote sensing images with complex backgrounds and extensive spatial variations [12], [13]. Specifically, CNNs perform poorly in capturing long-range dependencies and global contextual information due to their inherent local receptive fields and geometric architecture, which constrain their ability to capture broad spatial relationships. While spatial context modeling techniques such as CSFCN [14] and SRRNet [15] have enhanced spatial awareness through the introduction of cascaded pyramid pooling and hierarchical convolution, these methods typically focus on capturing homogeneous contextual dependencies and often overlook dynamic attribute relationships within the scene. This can lead to inaccurate contextual understanding, particularly in complex or confused remote sensing scene attributes. However, many existing information aggregation methods, such as graph convolutional networks (GCNs), still show limitations in handling homogeneity variations. While these methods excel in modeling local and global dependencies, they struggle to capture fine-grained attribute differences and significantly increase computational complexity, especially in remote sensing semantic understanding tasks involving complex backgrounds and diverse scenes. In contrast, our design of the attribute feature aggregation (AFA) module aims to

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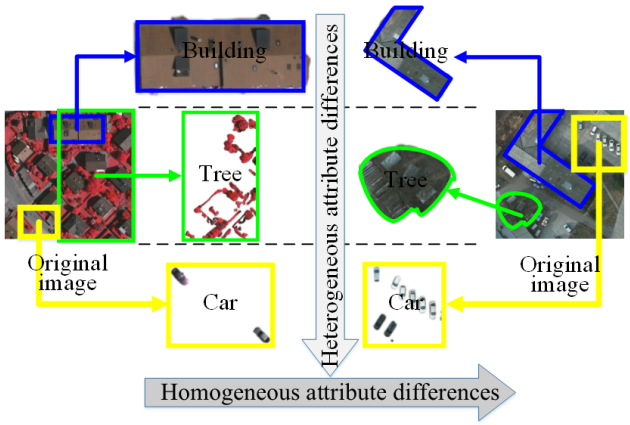


Fig. 1. Homogeneity and heterogeneity of differences between various attributes. Vertically, there are noticeable structural and spatial attribute differences between different attributes (building, tree, and car). Horizontally, when comparing the same attributes, there are evident homogeneous differences in the boundary structures of two types of buildings or two types of trees, whereas the homogeneous differences between two types of cars are less apparent.

enhance the model's sensitivity to subtle intraclass variations by fully integrating global and local contextual information with minimal computational overhead. This approach improves the model's performance in distinguishing nuanced differences within similar attributes, effectively addressing the challenges presented by homogeneous variations.

In remote sensing imagery, different land-cover categories such as buildings, trees, and cars exhibit significant structural and spatial attribute differences. For example, as shown in Fig. 1, there are evident heterogeneous differences between different types of land covers, which manifest in variations in shape, size, texture, and distribution. Within the same category of land covers, homogeneous differences exist, such as the differences in boundaries and shapes among various buildings, although they all fall under the category of buildings. Similarly, there are homogeneous differences among different trees and cars, although such differences may be less pronounced in certain categories (such as cars). These differences underscore the importance of capturing and modeling the relationships between different attributes. Current relational context modeling methods enhance the model's capability to handle heterogeneous scene attribute information by introducing attention mechanisms [16], [17], [18]—using pixel-level similarity calculations for weighted aggregation to achieve significant results in various remote sensing image analysis tasks—they primarily focus on interactions between pixels and often overlook the relationship between pixels and their surrounding global land-cover attributes. This leads to an underutilization of global background information (global distribution) and relative position prior information. This limitation is particularly prominent in remote sensing imagery [19], [20], which often covers extensive geographical areas containing complex features and backgrounds, along with notable homogeneity in attribute variations.

In response to the challenges highlighted, we propose a new framework for semantic understanding of remote sensing imagery, named comprehensive attribute difference attention network (CADANet). To fully learn and comprehend the impact of comprehensive attribute differences on the semantics of remote sensing imagery, CADANet employs a feature compression decoder (FCD) to reduce the dimensionality of feature representations to a more compact form. This process not only helps in reducing computational complexity but also diminishes the interference of homogeneous attribute differences in feature representations, thereby accentuating primary attribute characteristics. The AFA module in CADANet focuses on attribute-level global and local domain modeling, using fine-grained feature extraction and global contextual information to lessen the impact of homogeneous attribute differences. The AFA module includes two feature regressors that integrate context matrix information from both global and local domains, allowing the model to more accurately capture fine-grained differences within the same category. This enhancement significantly boosts the model's ability to distinguish between features of the same attribute. For learning and distinguishing heterogeneous attribute differences, we meticulously designed the context attribute-aware spatial attention (CAASA) module, which combines pixel-level global background information and relative positional information to capture long-distance dependencies using a self-attention mechanism, thus addressing heterogeneous attribute differences. Traditional fixed methods often overlook the relative positional relationships between pixels, while our CAASA module greatly enhances the model's sensitivity to spatial relationships between different attributes. The primary contributions of this article are as follows.

1) We propose a new network architecture, the CADANet, which focuses on addressing both homogeneous and heterogeneous attribute differences in remote sensing images. By integrating pixel-level interactions with attribute-level global–local contextual information, the model significantly enhances its capability to capture extensive spatial connections and contextual dependencies in remote sensing imagery.
2) For learning homogeneous attribute differences, we designed the AFA module, which combines fine-grained feature extraction with global context information. This enables the model to more accurately distinguish subtle variations within the same category, enhancing its ability to differentiate features with similar attributes.
3) The CAASA module combines pixel-level global background information and relative position priors, utilizing a self-attention mechanism to capture long-range dependencies and address heterogeneous attribute differences. This module greatly enhances the model's sensitivity to the spatial relationships between different classes of land cover, effectively reducing confusion between attributes.

II. Related Work

A. Convolution Methods for Semantic Understanding

The CNNs are widely employed in the semantic segmentation of remote sensing images, particularly adept at extracting local features. Recent studies have significantly enhanced the spatial perception capabilities of these models

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by introducing methods such as pyramid pooling and multiscale convolutions. However, these approaches tend to focus on capturing the contextual dependencies of homogeneous attributes, often overlooking the capture and modeling of dynamic association information within complex scenes [14], [15], [21], [22], [23], [24]. Another line of research attempts to capture the attribute differences between various semantics, especially at boundaries, through multilink feature aggregation. For instance, Wang et al. [25] employ a hierarchical architecture search method to set different boundary feature fusion structures for feature maps of varying resolutions. Wu et al. [26] significantly improve segmentation precision and edge clarity through multipath fusion and guided upsampling strategies. In addition, GBFNet [27] enhances feature extraction by integrating boundary information with spatial semantic context. Ni et al. [19] leverage edge information to guide the semantic segmentation process, achieving more precise segmentation by combining spatial information and semantic features. Despite these advancements, convolution-based methods still face limitations in handling long-distance dependencies and global contextual information, particularly when dealing with homogeneous attribute differences where they are susceptible to background noise interference. Works similar to our motivation, such as SCBANet [28], enhance model robustness against interference and the capability to handle class imbalance issues by incorporating semantic category assignment. This aligns with our approach of introducing the comprehensive AFA, which significantly reduces the impact of homogeneous attribute differences by integrating fine-grained feature extraction with global contextual information.

B. Transformer Methods for Semantic Understanding

The transformer model, due to its exceptional capability to capture long-distance dependencies, has progressively been applied in the semantic understanding of remote sensing images, achieving notable advancements [29], [30], [31]. DCSwin [17] captures multiscale and enhanced relational semantic features effectively by integrating the Swin transformer as both encoder and decoder. Although transformer models excel in capturing long-range dependencies, they face challenges in balancing local detail extraction with global dependencies. Liu et al. [18] introduced a global–local transformer segmentation framework that utilizes transformers within both the encoder and decoder to achieve consistent feature representation, thereby enhancing the precision of semantic understanding. Xu et al. [32] enhanced foreground prominence using transformers and attention mechanisms, effectively suppressing background noise and accentuating foreground targets for more precise segmentation in complex backgrounds. FTUnetFormer [33] proposed an attention-based method for remote sensing image semantic segmentation that significantly improves segmentation accuracy and edge clarity through the integration of object consistency loss and boundary preservation loss. In addition, EMIDI [20] introduced a method to enhance dataset-level and image-level contextual interactions by encouraging multilevel context interaction during feature extraction and fusion, but they primarily rely on global context information and lack sufficient modeling of fine-grained feature extraction and subtle differences within homogeneous attributes. Despite these transformer-based methods performing well in handling heterogeneous attribute differences, they often overlook the processing of local details and background noise. Furthermore, due to their relatively fixed attention mechanisms, they struggle to adapt flexibly to the dynamic changes in semantic attributes in remote sensing imagery.

C. Comprehensive Methods for Semantic Understanding

Hybrid methods combining CNNs and transformers are increasingly utilized in the semantic understanding of remote sensing images [29], [34], [35], [36], [37], [38], [39], [40]. A prevalent approach involves leveraging multiscale contextual information extraction and feature fusion to integrate the context data extracted by transformers with the local details captured by CNNs [12], [16], [41], [42]. Another line of research focuses on joint learning of features from different attributes or modalities. For instance, PACSCNet [43] enhances the semantic understanding of multimodal remote sensing images through cross-layer and cross-modal feature fusion. FTransUNet [44] achieves accurate semantic understanding by alternately merging multimodal features through a transformer architecture. BAnet [45] constructs texture and dependency pathways to extract fine-grained attribute features and long-distance dependency attributes, respectively, which are then integrated using a feature aggregation module to enhance understanding accuracy. In addition, IDRNet [46] introduces an intervention-driven relational network that simplifies attribute relation modeling by establishing pixel-level relations among attributes using diagnostic deletion methods. ABCNet [47] proposes an attention bilateral context network that effectively understands high-resolution remote sensing images by combining spatial and contextual pathways. These methods that integrate CNNs and transformers, by harnessing the strengths of both, effectively address the limitations when used separately. Transformers excel in capturing long-distance dependencies, while CNNs are proficient at extracting local detail features. By integrating both, these approaches demonstrate exceptional precision in handling complex tasks of semantic understanding in remote sensing imagery.

III. Methodology

The proposed CADANet architecture, as illustrated in Fig. 2, comprises three main components: the feature extraction backbone network, the AFA module, and the CAASA module. First, a ResNet-50 pretrained on ImageNet serves as the backbone network, extracting feature representations $\mathcal{F}$ from the input image. These features are then processed by the FCD to generate the initial probabilistic feature representation matrix $\mathcal{P}$. Subsequently, the AFA module takes both $\mathcal{F}$ and $\mathcal{P}$ as inputs to produce the global context information matrix $\mathcal{F}^{\mathcal{A}}$, which is reshaped into the global attribute feature representation matrix $\mathcal{F}_{\mathcal{G}}^{\mathcal{A}}$. Similarly, by reshaping $\mathcal{F}$ and $\mathcal{P}$, we obtain $\mathcal{F}_l$ and the initial probabilistic matrix $\mathcal{P}_l$. The AFA module then generates the local attribute feature representation

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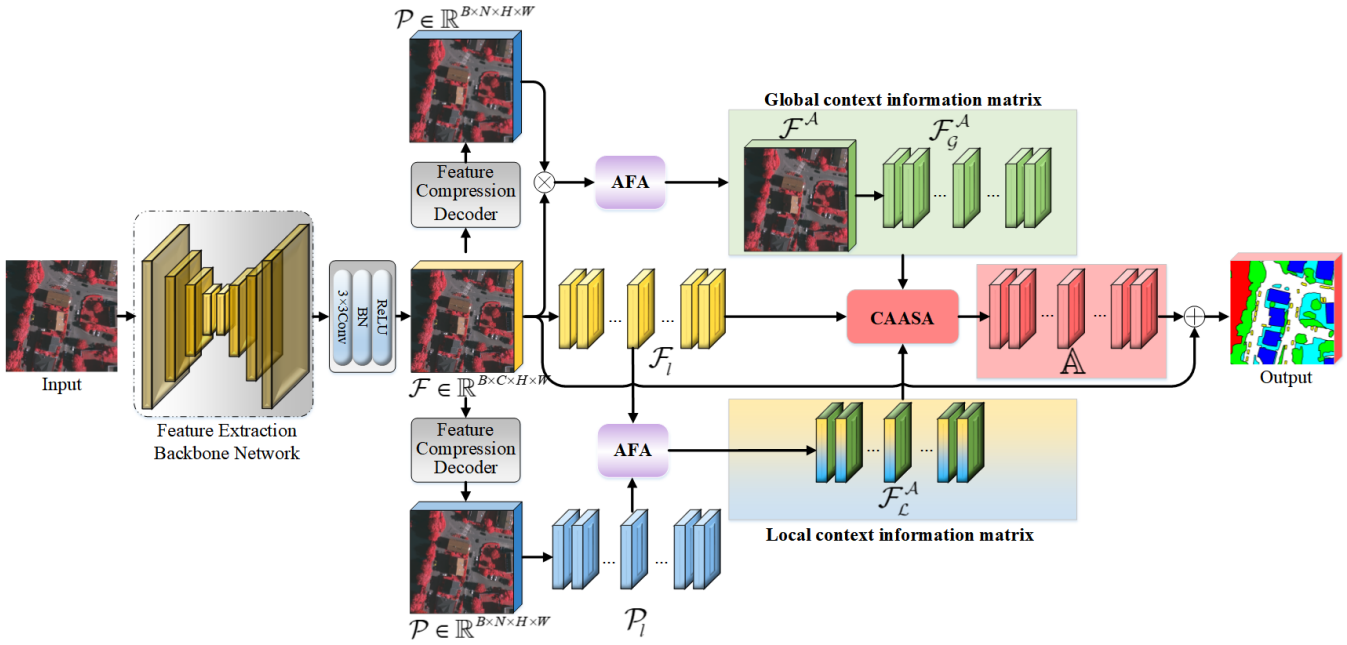


Fig. 2. Architecture of the CADANet.

matrix $\mathcal{F}_{\mathcal{L}}^{\mathcal{A}}$. In addition, the CAASA module uses $\mathcal{F}_l$, $\mathcal{F}_{\mathcal{L}}^{\mathcal{A}}$, and $\mathcal{F}_{\mathcal{G}}^{\mathcal{A}}$ as inputs to produce the enhanced feature representation $\mathbb{A}$. Finally, after restoring the original spatial dimensions, $\mathbb{A}$ is concatenated with $\mathcal{F}$ to obtain the final output feature representation.

A. Attribute Feature Aggregation Module

The AFA module of CADANet captures the relationships between pixels through global and local spatial modeling, incorporating attribute-level contextual information to mitigate background noise and attribute confusion. Our aim with the AFA is to ensure that the features of each pixel not only depend on the contextual information among similar pixels within local spatial blocks but also integrate the features of similar pixels at a global level, thereby reducing inconsistencies caused by homogeneous attribute differences. This approach enhances the consistency of the representation of similar pixels in the feature space, thereby diminishing the impact of homogeneous differences.

Initially, for the feature representation vector $\mathcal{F} \in \mathbb{R}^{B\times C\times H\times W}$ extracted from the backbone network, we deploy the FCD to generate a preliminary probability feature representation vector $\mathcal{P} \in \mathbb{R}^{B\times N\times H\times W}$ for the remote sensing image, where B represents the batch size, N represents the number of attributes, and $H \times W$ represents the height and width of the feature map, respectively. As shown in Fig. 3, the FCD first utilizes a 1×1 convolution layer to reduce the number of channels in the feature representation matrix from C to $C/2$. This reduction in feature dimensions highlights the primary features while simultaneously diminishing the influence of homogeneous attribute differences. Following this, the reduced feature map undergoes batch normalization to standardize the feature distribution across each channel, further mitigating the impact of homogeneous attribute differences. The model applies the ReLU activation function and dropout operations to enhance its nonlinear expression capability, better adapting to the complex homogeneous attribute differences present in remote sensing images. Finally, a 1×1 convolution layer is applied to convert the number of channels in the feature map to the target attribute dimension N, ensuring that the model generates outputs consistent with the target attributes. This step guarantees that the output of the model aligns with the number of target attributes, facilitating accurate semantic understanding.

In the global spatial domain modeling aspect of the AFA module, we aim to accurately model the relationship between pixels and their corresponding attribute context in remote sensing images. For this purpose, the probabilistic feature representation matrix $\mathcal{P}$ serves as the foundation for determining the probability that each pixel belongs to a specific attribute, thus laying the groundwork for further attribute-related context aggregation.

To compute these probabilities, we apply softmax normalization, as shown in the following equation:

$$\mathcal{P} = \frac{\exp\left(\textbf{scale} \cdot \mathcal{P}_{b,n,i}\right)}{\sum_{j=1}^{HW} \exp\left(\textbf{scale} \cdot \mathcal{P}_{b,n,j}\right)} \tag{1}$$

where $\mathcal{P}_{b,n,i}$ represents the softmax-normalized probability of pixel i in batch b belonging to attribute n. The **scale** parameter adjusts the distribution of softmax values to balance the impact of homogeneous attribute differences. This scaling ensures that the probabilities are sensitive to subtle variations in attributes across spatial regions, which is crucial in remote sensing image processing since certain land-cover classes may exhibit subtle but meaningful differences. The use of softmax in this context ensures that the sum of probabilities across all pixels is 1, allowing a clearer interpretation of the relationships between pixels and attributes. This normalization helps reduce

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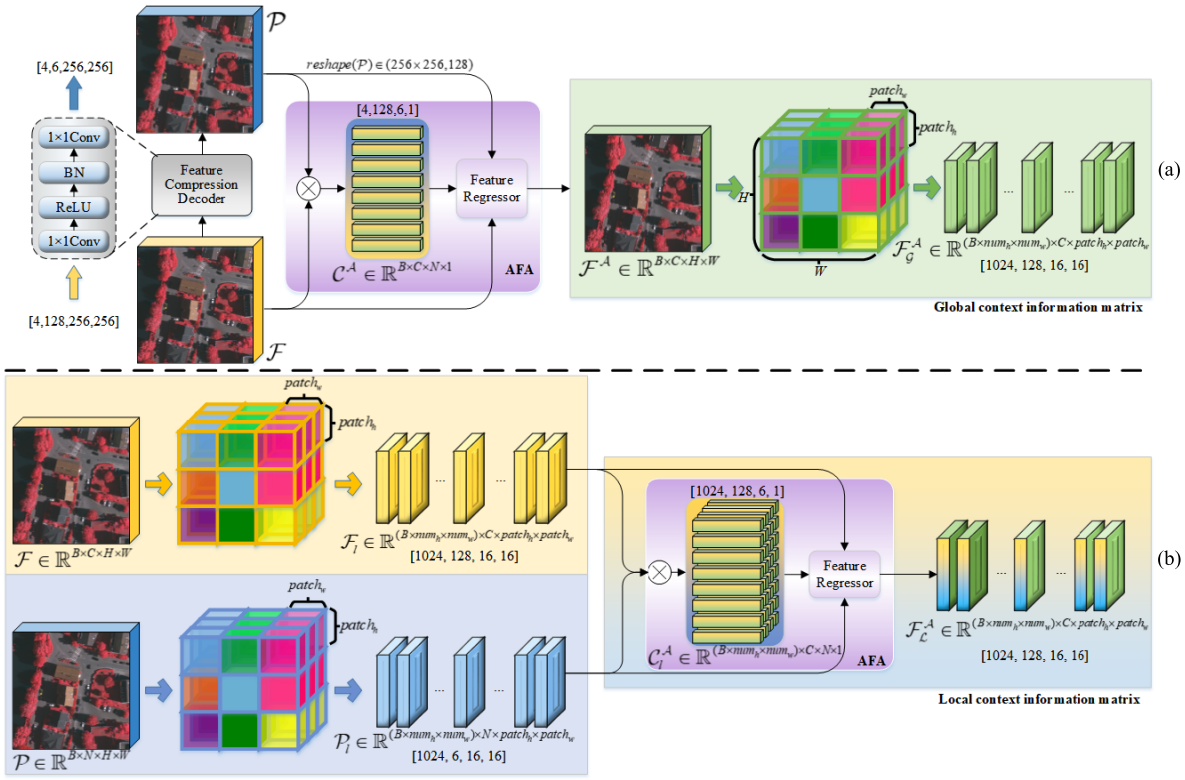


Fig. 3. Spatial domain modeling by the AFA module. (a) Global spatial domain modeling and (b) local spatial domain modeling.

the interference of background noise by assigning more weight to dominant attributes.

Next, we use these probabilities to compute the attribute-related context information for each attribute, as shown in the following equation:

$$\mathcal{C}^{\mathcal{A}} = \sum_{i=1}^{HW} (\mathcal{P}\mathcal{F}) \tag{2}$$

where $\mathcal{C}^{\mathcal{A}}$ is the aggregated attribute-related context. $\mathcal{F}$ represents the feature map, which reflects the original feature information extracted from the backbone network. The summation of all pixels effectively combines the raw features with the corresponding probabilities, producing a context vector influenced by both the feature intensity and the pixel attribute probability. This aggregation step is crucial for retaining spatial context information while focusing on attribute-specific details.

By combining the softmax-normalized probabilities with the feature map, we ensure that the resulting context is robust against noise while capturing the spatial relationships between pixels and their corresponding attributes.

Finally, the feature regressor ψ (see Algorithm 1) combines the attribute probability matrix $\mathcal{P}$, feature matrix $\mathcal{F}$, and global attribute context information $\mathcal{C}^{\mathcal{A}}$ across spatial locations to generate the new enhanced feature map $\mathcal{F}^{\mathcal{A}}$

$$\mathcal{F}^{\mathcal{A}} = \psi\left(\mathcal{P}, \mathcal{F}, \mathcal{C}^{\mathcal{A}}\right) \tag{3}$$

where $\mathcal{C}^{\mathcal{A}} \in \mathbb{R}^{B\times N\times C}$ represents the global attribute context obtained by weighted aggregation of all pixel and attribute information.

Regarding the design of ψ, where the resulting feature map contains the same features across all channels, our primary goal was to ensure that each pixel's features are fully integrated with attribute-related contextual information. This helps in reducing the interference from background noise and allows the model to better learn homogeneous attribute differences. By mapping the category context features to each channel, we aim to guarantee that the features at each pixel location are consistent with the corresponding attribute context.

For the obtained feature map, it is split along the spatial dimension to calculate the number of patches that can be segmented along each dimension, num_h and num_w. This process reshapes and results in the formation of a global context information matrix $\mathcal{F}_{\mathcal{G}}^{\mathcal{A}} \in \mathbb{R}^{(B\times \text{num}_h\times \text{num}_w)\times C\times \text{patch}_h\times \text{patch}_w}$ corresponding to the input remote sensing image.

By segmenting along the spatial dimension, the homogeneous attribute differences are fully considered and modeled within different blocks

$$\text{num}_h = \frac{H}{\text{patch}_h}, \quad \text{num}_w = \frac{W}{\text{patch}_w} \tag{4}$$

where patch_h and patch_w, respectively, represent the height and width dimensions of the segmented patches.

For local spatial domain modeling, we first reshape the feature representation vector $\mathcal{F}$ and the probability feature representation vector $\mathcal{P}$ using the same dimensional splitting method, dividing the large feature map into multiple small patches. Each patch can be processed independently, thereby enhancing the model's ability to capture and process local details.

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Algorithm 1 Pseudo Code of Feature Regressor (ψ)

Input: feature representation vector $\mathcal{F} \in \mathbb{R}^{C\times H\times W}$, probabilistic feature representation vector $\mathcal{P} \in \mathbb{R}^{B\times N\times H\times W}$, contextual information matrix $\mathcal{C}^{\mathcal{A}} \in \mathbb{R}^{\mathrm{B}\times\mathrm{C}\times\mathrm{N}}$

Output: attribute contextual information matrix $\mathcal{F}^{\mathcal{A}} \in \mathbb{R}^{B\times C\times H\times W}$

```
//Perform argmax on the P to get the predicted attribute P^L ∈ R^{B×H×W}
for b in 0 ··· (B − 1) do
| for i in 0 ··· (H − 1) do
| | for j in 0 ··· (W − 1) do
| | | P^L[b, i, j] = arg max(P[b, :, i, j])
| | end for
| end for
end for
Reshape C^A to [B, N, C] and initialize the new context information F^A ∈ R^{B×H×W×C}
//for each batch b and each pixel position (i, j)
for b in 0 ··· (B − 1) do
| for i in 0 ··· (H − 1) do
| | for j in 0 ··· (W − 1) do
| | | attribute_idx = P^L[b, i, j]
| | | for c in 0 ··· (C^A − 1) do
| | | | F^A[b, i, j, c] = context[b, attribute_idx, c]
| | | end for
| | end for
| end for
end for
Reshape the attribute contextual information matrix F^A to [B, C, H, W]
Return F^A
```

After splitting and reshaping, we obtain $\mathcal{F}_l \in \mathbb{R}^{(B\times \mathrm{num}_h\times \mathrm{num}_w)\times C\times \mathrm{patch}_h\times \mathrm{patch}_w}$ and $\mathcal{P}_l \in \mathbb{R}^{(B\times \mathrm{num}_h\times \mathrm{num}_w)\times N\times \mathrm{patch}_h\times \mathrm{patch}_w}$. Next, we use the AFA module to extract local context information by calculating attribute-related features within each patch, resulting in enhanced context representations. This process fully accounts for the impact of homomorphic attribute differences in different local patches. Finally, the local category probability $\mathcal{P}_l$, local features $\mathcal{F}_l$, and local context information $\mathcal{C}_l^{\mathcal{A}}$ (where $\mathcal{C}_l^{\mathcal{A}}$ follows the same principle as $\mathcal{C}^{\mathcal{A}}$) are fused through the feature regressor ψ, generating the enhanced local feature map $\mathcal{F}_{\mathcal{L}}^{\mathcal{A}} \in \mathbb{R}^{(B\times \mathrm{num}_h\times \mathrm{num}_w)\times C\times \mathrm{patch}_h\times \mathrm{patch}_w}$

$$\mathcal{F}_{\mathcal{L}}^{\mathcal{A}} = \psi\left(\mathcal{P}_l, \mathcal{F}_l, \mathcal{C}_l^{\mathcal{A}}\right). \tag{5}$$

B. Context Attribute-Aware Spatial Attention Module

In addressing heterogeneous attribute differences in remote sensing imagery, the CAASA module (as shown in Fig. 4) enhances the model's sensitivity to spatial relationships between different land-cover attributes by integrating global and local contextual information, along with pixel-level global background and relative positional priors. Due to the spatial interdependence of different land-cover types—for example, wetlands typically occur near rivers—the CAASA module utilizes scene awareness to capture and model these spatial distribution patterns, effectively distinguishing differences from other land-cover types. This approach enables the model to more accurately comprehend the spatial distribution patterns of land cover, thereby reducing attribute confusion. Moreover, the CAASA module further improves the model's capability to capture long-range dependencies through a self-attention mechanism. This is particularly crucial in remote sensing images, where many land-cover structures span large areas; effective capture of long-distance dependencies among pixels is essential for better recognition and differentiation of land-cover types with heterogeneous attribute differences. The self-attention mechanism allows the model to identify potential spatial associations between pixels, especially in scenes with complex heterogeneous structures. Finally, by embedding a pixel-level scene attribute-aware spatial attention mechanism, combined with global background information and relative positional priors, the CAASA module further enhances the model's accuracy in land-cover recognition. This module not only captures local features but also integrates global contextual information, effectively reducing confusion among attributes.

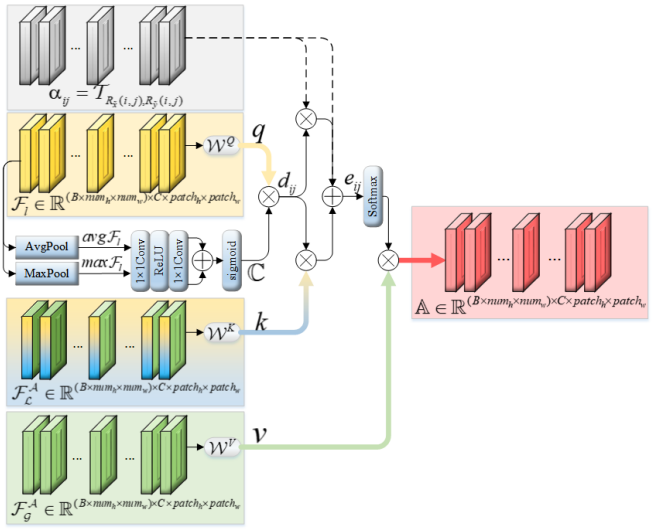


Fig. 4. Architecture of the CAASA module.

By capturing the relative positional relationships between pixels in the image, we enhance the model's understanding of spatial structure information. Combining the feature matrix obtained from the AFA module, we redefine the query, key, and value in the attention mechanism using the linear transformation matrices $\mathcal{W}^Q$, $\mathcal{W}^K$, and $\mathcal{W}^V$. Specifically, $X_Q = \mathcal{F}_l$, $X_K = \mathcal{F}_{\mathcal{L}}^{\mathcal{A}}$, and $X_V = \mathcal{F}_{\mathcal{G}}^{\mathcal{A}}$

$$\begin{aligned} q &= \mathcal{F}_l\mathcal{W}^Q \\ k &= \mathcal{F}_{\mathcal{L}}^{\mathcal{A}}\mathcal{W}^K \\ v &= \mathcal{F}_{\mathcal{G}}^{\mathcal{A}}\mathcal{W}^V. \end{aligned} \tag{6}$$

We use the partition function par() to map the relative positional offsets of pixels to a finite set of integers. This approach reduces computational complexity and the number

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of training parameters. The specific formula is as follows:

$$\mathrm{par}(x) = \begin{cases} [x], & |x| \le \mu \\ \mathrm{sign}(x) \times \min\left(\lambda, \left[\mu + \dfrac{\ln\left(\frac{|x|}{\mu}\right)}{\ln\left(\frac{\nu}{\mu}\right)}\right]\right)(\lambda - \mu), & |x| > \mu \end{cases} \tag{7}$$

where $[\cdot]$ denotes the rounding operation, μ determines the segmentation points, λ controls the range of the output, and ν adjusts the curvature of the logarithmic part. For a given position i and j, the relative displacement $R()$ in the horizontal and vertical directions is calculated as follows:

$$\begin{aligned} R_{\tilde{x}}(i, j) &= \mathrm{par}\left(\tilde{x}_i - \tilde{x}_j\right) \\ R_{\tilde{y}}(i, j) &= \mathrm{par}\left(\tilde{y}_i - \tilde{y}_j\right). \end{aligned} \tag{8}$$

Using learnable scalars $\mathcal{T}_{R_{\tilde{x}}(i,j), R_{\tilde{y}}(i,j)}$, we dynamically adjust the relationship weights between positions to better capture the spatial structure information within the input features. The relative positional encoding matrix α_{ij} is retrieved from a lookup table

$$\alpha_{ij} = \mathcal{T}_{R_{\tilde{x}}(i,j), R_{\tilde{y}}(i,j)}. \tag{9}$$

Specifically, we first perform adaptive average pooling and max pooling on the input feature matrix $\mathcal{F}_l$, obtaining the global feature representation for each channel. This step helps address heterogeneous attribute differences by extracting global information and facilitating differentiated processing

$$\begin{aligned} \mathrm{avg}\mathcal{F}_l &= \mathrm{AvgPool}(\mathcal{F}_l) \\ \max \mathcal{F}_l &= \mathrm{MaxPool}(\mathcal{F}_l). \end{aligned} \tag{10}$$

Next, the feature representations $\mathrm{avg}\mathcal{F}_l$ and $\max\mathcal{F}_l$ are processed through two 1×1 convolution layers to compute the importance scores for each channel. These scores reflect the heterogeneous attribute differences between different categories, aiding the model in accurately distinguishing between various attributes. Finally, by applying a sigmoid nonlinear transformation to these scores, we generate the channel context encoding matrix $\mathbb{C}$

$$\begin{aligned} \mathbb{C} = \sigma(&\mathrm{conv}(\mathrm{Relu}(\mathrm{conv}(\mathrm{avg}\mathcal{F}_l))) \\ &+ \mathrm{conv}(\mathrm{Relu}(\mathrm{conv}(\max\mathcal{F}_l)))). \end{aligned} \tag{11}$$

The transformed query vector q is adjusted to obtain the feature query matrix d_{ij}, which incorporates channel context and remote sensing semantic information

$$d_{ij} = q \otimes \mathbb{C}^{\mathrm{T}}. \tag{12}$$

In the CAASA module, we integrate various global background and position encoding information by introducing the relative position encoding matrix α_{ij} to interact with the updated query d_{ij} and the key feature matrix. This interaction captures more contextual information by dynamically adjusting the encoding according to the changes in the input features. The attention weight e_{ij} between the query q_i and the key k_j is calculated through this interaction

$$e_{ij} = \frac{d_{ij}\alpha_{ij}^{\mathrm{T}} + d_{ij}k^{\mathrm{T}} + \alpha_{ij}}{\sqrt{d}} \tag{13}$$

where $\sqrt{d}$ represents a normalization factor used to prevent the attention scores from becoming too large, ensuring stability and convergence during model training. The attention score e_{ij} not only considers the fundamental relationship between the query and the key but also integrates relative position encoding and contextual information. This enables the model to more comprehensively understand the spatial and global background relationships between pixels in the image. Finally, we apply the softmax-normalized attention weights e_{ij} to the value features v_j at all positions jth in the input feature map, resulting in the weighted sum feature matrix $\mathbb{A}_i$

$$\mathbb{A}_i = \sum_{j=1}^{n} \left(\frac{\exp(e_{ij})}{\sum_{k=1}^{n}\exp(e_{ik})}\right) v_j. \tag{14}$$

IV. Experiments

This section will cover three subsections pertaining to preliminary work for the experiment. The first part demonstrates the dataset used in the experiment. The second subsection describes the evaluation metrics in detail. The third parts introduce the baseline models used for comparison in this study.

This section will conduct some experiments to test the proposed method. First, we evaluate the proposed modules using ablation experiments. Next, we compare the entire model with some recent state-of-the-art methods. Finally, we conduct some visualization experiments to analyze the operating principles of some modules qualitatively.

A. Materials and Data Preparation

1) The ISPRS Vaihingen dataset, provided by the International Society for Photogrammetry and Remote Sensing (ISPRS), is a high-resolution remote sensing image dataset extensively used for semantic segmentation and land-cover classification tasks. This dataset consists of 33 high-resolution true orthophoto (TOP) tile images and corresponding digital surface models (DSMs), with the DSM channel being optionally excluded in some experiments. It defines six land-cover labels: impervious surfaces, buildings, low vegetation, trees, cars, and clutter. These images are characterized by extremely high resolution, capturing fine details of urban environments and providing rich spatial and spectral information. For evaluating model performance, we used 16 of the 33 images for training and validation, with 11 images (IDs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, and 21) for training and five images (IDs: 23, 25, 27, 29, and 31) for validation. The remaining images were used for testing. Due to its high-resolution and complex urban scenes, this dataset is widely used to test and validate various advanced methods for semantic understanding of remote

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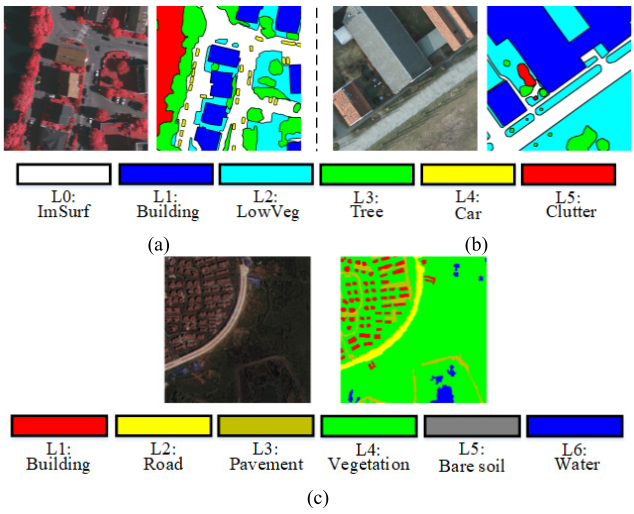


Fig. 5. Experimental samples and their corresponding attribute labels. (a) Sourced from the ISPRS Vaihingen dataset, (b) sourced from the ISPRS Potsdam dataset, and (c) sourced from the WHDLD dataset.

sensing images, making it a standard benchmark for assessing model performance.

2) The ISPRS Potsdam dataset is another high-resolution remote sensing image dataset provided by the ISPRS, specifically designed for semantic segmentation and land-cover classification tasks. This dataset comprises 38 TOP tile images, each with a resolution of 5 cm. Similar to the ISPRS Vaihingen dataset, the Potsdam dataset includes multiple categories of land-cover labels such as impervious surfaces, buildings, low vegetation, trees, cars, and water. These images cover a larger urban area, featuring more complex and diverse land-cover types. We selected 24 out of the 38 images for training and validation, with the following image IDs: 2_10, 2_11, 2_12, 3_10, 3_11, 3_12, 4_10, 4_11, 4_12, 5_10, 5_11, 5_12, 6_7, 6_8, 6_9, 6_10, 6_11, 6_12, 7_7, 7_8, 7_9, 7_10, 7_11, and 7_12. The remaining images were used for testing.

3) The WHDLD dataset [48] is a high-resolution remote sensing dataset specifically designed for semantic segmentation and land-cover classification tasks. This dataset encompasses multiple representative world heritage sites around the globe, featuring complex terrains and diverse land-cover types, and providing rich spatial and spectral information. The WHDLD dataset offers multispectral, high-resolution remote sensing imagery, with each image boasting submeter resolution, allowing for precise capture of fine details in urban and natural environments. The dataset includes a total of 4090 images, annotated with six land-cover types: bare soil, building, pavement, road, vegetation, and water, as shown in Fig. 5(c). To train and evaluate model performance, the dataset is randomly divided into training, validation, and test sets, comprising 80%, 10%, and 10% of the total images, respectively.

B. Evaluation Metrics

In our study, we utilize several key evaluation metrics, including mean accuracy (mAcc), mean F1 score (mF1), and mean intersection over union (mIoU)

$$P_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i} \quad R_i = \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FN}_i} \tag{15}$$

$$\mathrm{F1} = \frac{1}{n}\sum_{i=1}^{n} 2 \times \frac{P_i \times R_i}{P_i + R_i} \tag{16}$$

$$\mathrm{mIoU} = \frac{1}{n}\sum_{i=1}^{n} \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i} \tag{17}$$

$$\mathrm{mAcc} = \frac{1}{n}\sum_{i=1}^{n} \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FN}_i} \tag{18}$$

where n denotes the number of attributes, TP_i represents the true positives for attribute i, FP_i denotes the false positives for attribute i, and FN_i indicates the false negatives for attribute i. By employing these metrics, we can comprehensively evaluate the model's performance under various conditions, enabling us to select the optimal semantic understanding algorithm.

C. Implementation Details

During training, we used a single A100 GPU for 180 epochs, taking approximately 20 h, while inference without image visualization took around 4 min. For model optimization, we employed the AdamW optimizer with a weight decay (L2 regularization) set to $1E^{-2}$. The initial learning rate was set to $2E^{-4}$ and dynamically adjusted using a cosine annealing scheduler. To quantitatively evaluate model performance, we used a cross-entropy loss function combined with backpropagation to optimize model parameters. The cross-entropy loss effectively guides the model to gradually minimize errors between predicted segmentation maps and ground truth labels, enhancing segmentation accuracy. This approach ensures efficient semantic understanding across varying scales, orientations, and complex scenarios. Each patch had an aspect ratio of 16 × 16. To further enhance model generalization and robustness, we divided the original remote sensing images into smaller 512 × 512 pixel patches. During training, we applied a series of data augmentation techniques, including random scaling (range 0.5–1.5), random vertical flips, horizontal flips, and random rotations.

D. Ablation Study

We conducted ablation experiments on the FCD module, AFA module, and CAASA module to validate the necessity and performance improvement contributed by each component. The experiments were performed on the ISPRS Vaihingen and ISPRS Potsdam datasets, with results shown in Table I.

1) Effectiveness of the AFA Module: The AFA module is designed to effectively model attribute-level context in both global and local spatial domains, thereby mitigating the impact of homomorphic attribute differences. As observed in Table I, the performance of the model on the ISPRS Vaihingen dataset significantly improves with the inclusion of the AFA module, particularly when incorporating both global and local domains. For instance, when using FCD, AFA (global and local), and the classic attention (CA) mechanism, the mIoU reaches 83.17%,

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TABLE I

ABLATION STUDY RESULTS (%) WITH DIFFERENT MODULES ON TWO DATASETS. THE BOLD MEANS THE BEST PERFORMANCE

Dataset	FCD	AFA		CAASA	CA	mIoU	mAcc	mF1
		global	local					
ISPRS Vaihingen	✓	✓	×	×	✓	82.45	91.68	89.60
	✓	×	✓	×	✓	82.03	90.99	89.18
	✓	✓	✓	×	✓	83.17	92.74	90.47
	✓	×	×	✓	×	83.66	93.10	90.87
	✓	✓	✓	✓	×	**84.91**	**93.62**	**91.69**
ISPRS Potsdam	✓	✓	×	×	✓	85.16	90.12	91.09
	✓	×	✓	×	✓	84.67	89.44	90.72
	✓	✓	✓	×	✓	85.83	91.15	91.95
	✓	×	×	✓	×	86.38	91.49	92.36
	✓	✓	✓	✓	×	**87.59**	**92.04**	**93.20**
WHDLD	✓	✓	×	×	✓	62.35	83.30	75.45
	✓	×	✓	×	✓	61.86	82.62	75.04
	✓	✓	✓	×	✓	63.03	84.33	76.28
	✓	×	×	✓	×	63.55	84.70	76.73
	✓	✓	✓	✓	×	**64.77**	**85.24**	**77.55**

TABLE II

ABLATION STUDY RESULTS (%) OF DIFFERENT STRUCTURES IN THE CAASA MODULE. THE BOLD MEANS THE BEST PERFORMANCE

Dataset	CAASA			mIoU	mAcc	mF1
	α_{ij}	$\mathbb{C}$	q, k, v			
ISPRS Vaihingen	×	×	✓	83.17	92.74	90.47
	✓	×	✓	84.55	93.10	91.38
	×	✓	✓	83.90	92.83	90.95
	✓	✓	✓	**84.91**	**93.62**	**91.69**
ISPRS Potsdam	×	×	✓	85.83	91.15	91.95
	✓	×	✓	87.23	91.52	92.89
	×	✓	✓	86.58	91.25	92.46
	✓	✓	✓	**87.59**	**92.04**	**93.20**
WHDLD	×	×	✓	63.02	84.34	76.30
	✓	×	✓	64.42	84.73	77.26
	×	✓	✓	63.75	84.46	76.81
	✓	✓	✓	**64.77**	**85.24**	**77.55**

mAcc improves to 92.74%, and mF1 achieves 90.47%. Without considering global and local domains, the mIoU decreases by 0.72% and 1.14%, mAcc drops by 1.06% and 1.75%, and mF1 reduces by 0.87% and 1.29%, respectively. These results indicate that the AFA module, by integrating global and local contextual information, effectively captures fine-grained features and contextual dependencies within the same attribute, thereby significantly enhancing segmentation accuracy and precision.

The effectiveness of the AFA module is further validated on the ISPRS Potsdam dataset. When using only FCD, the global AFA module, and CA, the mIoU is 85.16%, mAcc is 90.12%, and mF1 is 91.09%. With the addition of the AFA (global and local) and CA modules, the mIoU increases to 85.83%, mAcc to 91.15%, and mF1 to 91.95%. This improvement demonstrates the critical role of the AFA module in global and local attribute-level context modeling, effectively reducing the impact of homomorphic attribute differences through fine-grained feature extraction and global context integration, thereby enhancing the model's adaptability to complex scenes.

For the WHDLD dataset, the ablation study further validates the robustness and adaptability of the AFA module. The WHDLD dataset presents a high level of complexity in remote sensing image processing tasks, encompassing a wide variety of land-cover types. When the AFA module is not employed, the model achieves an mIoU of only 63.55%, mAcc of 84.70%, and mF1 of 76.73%. However, with the incorporation of the AFA module, leveraging both global and local domains, the model's mIoU significantly improves to 64.77%, mAcc rises to 85.24%, and mF1 increases to 77.55%. This improvement indicates that the AFA module, by integrating global and local contextual information, effectively enhances the model's ability to recognize and understand diverse land-cover types and complex scenes.

2) Effectiveness of the CAASA Module: The CAASA module aims to combine pixel-level global background information and relative position prior information to capture long-range dependencies through a self-attention mechanism, addressing the issue of heterogeneous attribute differences. On the ISPRS Vaihingen dataset, using FCD, AFA (global and local), and CA, the model achieves mIoU, mAcc, and mF1 of 83.17%, 92.74%, and 90.47%, respectively. However, as shown in Table I, when the CAASA module replaces the CA module, the model performance further improves, with mIoU, mAcc, and mF1 increasing by 1.74%, 0.88%, and 1.22%, respectively. This indicates that the CAASA module effectively captures long-range dependencies among pixels by integrating global background information and relative position priors, thereby reducing confusion among heterogeneous attribute differences and significantly enhancing segmentation accuracy and robustness.

The results on the ISPRS Potsdam dataset similarly validate the effectiveness of the CAASA module. Using FCD, AFA (global and local), and CA, the mIoU, mAcc, and mF1 are 85.83%, 91.15%, and 91.95%, respectively. When the CAASA module is introduced, all metrics show significant improvement, particularly with mIoU and mF1 increasing by 1.76% and 1.25%, respectively. This further proves the advantages of the CAASA module in capturing long-range dependencies and handling heterogeneous attribute differences, utilizing a self-attention mechanism and comprehensive contextual information, thereby significantly enhancing segmentation performance in complex backgrounds.

For the WHDLD dataset, the effectiveness of the CAASA module has also been fully validated. Compared to using the CA module, incorporating the CAASA module under the condition that other modules are used normally results in an improvement of 1.74, 0.91, and 1.27 in mIoU, mAcc, and mF1, respectively. This significant performance improvement indicates that the CAASA module, by integrating global background information and relative positional priors, effectively captures long-range dependencies between different land-cover types in remote sensing images, reduces the interference from heterogeneous attribute differences, and significantly enhances the model's interpretive accuracy and robustness.

When evaluating the effectiveness of the CAASA module, we focused on the roles of different structural components, including α_{ij}, $\mathbb{C}$, and the CA mechanism (q, k, v). The specific ablation experiment results are shown in Table II. Essentially,

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using only (q, k, v) equates to the CA. As demonstrated in Table I, although CA improves model performance in certain cases, it is limited compared to the performance achieved with the CAASA module, indicating that traditional attention mechanisms alone are insufficient for fully capturing long-range dependencies and contextual information in complex scenes.

In the ISPRS Vaihingen dataset, the introduction of α_{ij} led to an increase in the mIoU from 83.17% to 84.55%, mAcc from 92.74% to 93.10%, and mF1 from 90.47% to 91.38%. This demonstrates that α_{ij} effectively captures long-range dependencies between pixels, enhancing the model's fine-grained understanding of object boundaries in complex remote sensing images. When the channel context information $\mathbb{C}$ was introduced independently, performance improved across all metrics, though the gains were slightly less than those achieved by α_{ij}. This suggests that while channel context aids the model in capturing global background information, it is less effective than α_{ij} in integrating long-range dependencies between complex features. When α_{ij} and $\mathbb{C}$ were combined, performance further improved, with the mIoU reaching 84.91%, mAcc at 93.62%, and mF1 at 91.69%. This indicates that the integration of α_{ij} and $\mathbb{C}$ enhances the model's global semantic understanding and its ability to capture more detailed contextual information.

A similar trend was observed in the ISPRS Potsdam dataset. After incorporating α_{ij} and $\mathbb{C}$, the mIoU increased from 85.83% to 87.59%, mAcc from 91.15% to 92.04%, and mF1 from 91.95% to 93.20%. This highlights that in complex urban environments, the combination of α_{ij} and $\mathbb{C}$ significantly boosts segmentation accuracy, particularly when dealing with high-resolution features such as buildings and roads, where the model demonstrated enhanced robustness.

The WHDLD dataset showed similarly impressive results. The integration of α_{ij} and $\mathbb{C}$ improved the mIoU from 63.02% to 64.77%, mAcc from 84.34% to 85.24%, and mF1 from 76.30% to 77.55%. This indicates that even in datasets characterized by substantial heterogeneity in land-cover attributes, the combination of α_{ij} and $\mathbb{C}$ effectively enhances the model's scene understanding. The CAASA module, by capturing long-range dependencies and global background information, improves the handling of complex backgrounds and heterogeneous attribute differences.

These significant improvements further validate that the combination of α_{ij} and $\mathbb{C}$ is highly effective in capturing long-range dependencies and global context information between objects in remote sensing images. This is particularly important for datasets encompassing diverse and complex land-cover types, demonstrating that the CAASA module can effectively manage heterogeneous attribute differences by integrating self-attention mechanisms with global background information and relative positional priors, thereby enhancing the model's robustness and semantic understanding in challenging environments.

3) Sensitivity Analysis of Patch Size: Table III presents the ablation study results for CADANet's semantic understanding performance on the ISPRS Vaihingen dataset with varying patch sizes. The experiments demonstrate that as patch size increases from 4 × 4 to 16 × 16, the model's mIoU, mAcc, and mF1 scores steadily improve, reaching optimal performance at a 16 × 16 patch size. At this configuration, CADANet achieved an mIoU of 84.91%, an mAcc of 93.62%, and an mF1 of 91.69%. This improvement suggests that an intermediate patch size effectively balances global and local information extraction, enabling the model to capture complex land-cover relationships within remote sensing images.

TABLE III
ABLATION STUDY RESULTS (%) WITH DIFFERENT PATCH SIZE ON THE ISPRS VAIHINGEN DATASETS. THE BOLD MEANS THE BEST PERFORMANCE

Patch Size	mIoU	mAcc	mF1
4×4	83.02	91.06	90.04
8×8	84.15	93.19	90.82
16×16	**84.91**	**93.62**	**91.69**
32×32	84.45	93.54	91.27

However, when the patch size further increases to 32 × 32, performance metrics slightly decline. This indicates that while larger patches enhance the model's ability to capture long-range dependencies, an excessive patch size can compromise local detail, reducing accuracy in recognizing small-scale features.

E. Contrast Evaluation for the State-of-the-Art Methods

We conducted a comprehensive comparison of our proposed method with several state-of-the-art semantic understanding methods for remote sensing images developed in recent years. These methods encompass advanced techniques that combine transformer and CNN, such as Swin-SASPP [12], CMTFNet [16], DCSwin [17], GLOTS [18], FTUnetFormer [33], CMGFNet [41], UNetFormer [42], FTransUNet [44], and P-Swin [49]. In addition, we included commonly used multiscale feature fusion methods, such as EIGNet [19], EMIDI [20], MFNet [26], GBFNet [27], SCBANet [28], RSSFormer [32], UNetFormer [42], PACSCNet [43], BAnet [45], IDRNet [46], ABCNet [47], MGNet [50], BIBED-Seg [51], and CTMU-Net [52]. These methods have demonstrated outstanding performance in multiscale feature extraction, fusion, and enhancement, significantly improving the understanding of semantic information in remote sensing images. By integrating multiscale information and jointly learning different attribute features, these methods excel at capturing the rich and complex semantic features in remote sensing images, providing robust support for enhancing the accuracy and robustness of semantic understanding in remote sensing imagery.

As shown in Table IV, on the ISPRS Vaihingen dataset, CADANet achieved IoUs of 94.62%, 93.77%, 75.03%, 78.12%, and 83.03% for impervious surfaces, buildings, low vegetation, trees, and cars, respectively. These results indicate that CADANet exhibits high semantic understanding accuracy across various attributes, especially outperforming other methods in building and car attributes. Overall, CADANet achieved an mIoU of 84.91%, mAcc of 93.62%, and mF1 of 91.69%, surpassing all other comparison models. For instance, Swin-SASPP achieved an mIoU of 68.79%, while RSSFormer achieved an mIoU of 70.07%. These traditional single-model methods have limitations in capturing both global

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TABLE IV

COMPARISON EXPERIMENT RESULTS (%) WITH STATE-OF-THE-ART METHODS ON THE ISPRS VAIHINGEN DATASET. THE BOLD MEANS THE BEST PERFORMANCE. UNDERLINE REPRESENTS SECOND BEST PERFORMANCE

Method	IoU					mIoU(%)	mAcc(%)	mF1(%)
	ImSurf	Building	LowVeg	Tree	Car			
Swin-SASPP [12]	81.66	89.32	67.59	75.86	63.88	68.79	76.84	-
IDRNet[46]	80.48	87.51	65.34	75.43	64.61	68.98	-	-
CMTFNet[16]	80.21	86.98	65.73	75.63	63.38	69.25	-	-
BIBED-Seg[51]	70.60	76.10	63.0	67.80	62.30	69.60	87.70	-
GBFNet[27]	80.75	86.05	68.19	76.47	67.00	69.89	87.07	-
RSSFormer[32]	80.50	87.55	65.48	75.61	67.22	70.07	-	-
GLOTS[18]	82.65	89.17	69.36	77.04	67.55	70.13	77.81	-
SCBANet[28]	-	-	-	-	-	70.76	85.07	82.45
EMIDI[20]	80.94	88.48	66.74	75.72	67.50	71.50	-	-
MGNet[50]	-	-	-	-	-	81.15	88.74	89.11
ABCNet[47]	93.29	91.18	73.41	76.00	72.36	81.25	92.75	89.38
PACSCNet[43]	86.06	92.16	68.59	**78.97**	85.46	82.25	89.95	-
CMGFNet[41]	-	-	-	-	-	82.26	-	90.00
EIGNet[19]	-	-	-	-	-	83.05	91.21	90.61
BAnet[45]	92.96	91.25	74.01	76.39	81.16	83.15	92.82	90.61
UNetFormer[42]	94.35	93.18	74.67	77.10	79.99	83.56	93.52	91.00
DCSwin[17]	94.38	93.63	74.61	76.43	81.50	84.11	93.53	91.15
FTransUNet[44]	-	-	-	-	-	84.23	-	91.21
FTUnetFormer[33]	94.31	93.31	74.64	77.25	82.33	84.37	93.53	91.31
CADANet(our)	**94.62**	**93.77**	**75.03**	78.12	**83.03**	**84.91**	**93.62**	**91.69**

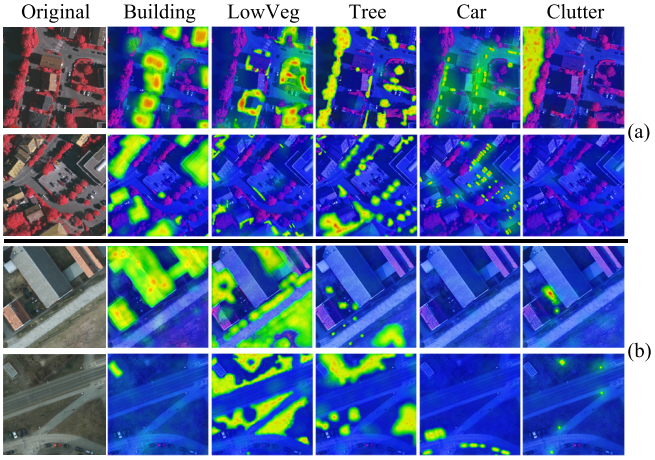


Fig. 6. Visualization of the class-specific activation map on example images. (a) Results on the ISPRS Vaihingen dataset and (b) results on the ISPRS Potsdam dataset.

and local information. While transformer-based architectures such as GLOTS and EMIDI perform well in handling long-range dependencies, their mIoUs of 70.13% and 71.50%, respectively, fell short of our model's performance in complex background scenarios. This further demonstrates that by integrating global background information and relative positional priors, CADANet significantly enhances the model's capability to handle heterogeneous attributes, long-range dependencies, and complex background information. Fig. 6(a) shows class-specific activation maps generated by CADANet on the ISPRS Vaihingen and ISPRS Potsdam datasets, clearly illustrating CADANet's superior performance in capturing high-quality activation maps for different attributes, such as buildings, low vegetation, trees, cars, and clutter. Fig. 7 further illustrates the visualization results of different models on the ISPRS Vaihingen dataset. It is evident that CADANet outperforms other models across various categories, especially in complex scenes involving buildings and vegetation.

In the ISPRS Potsdam dataset (as shown in Table V), CADANet continued to exhibit its strong semantic understanding capabilities, achieving IoUs of 89.33%, 94.26%, 79.01%, 81.87%, and 93.46% for impervious surfaces, buildings, low vegetation, trees, and cars, respectively. Overall, CADANet achieved an mIoU of 87.59%, mAcc of 92.04%, and mF1 of 93.20%. These results reaffirm CADANet's superior performance in complex urban environments. In this dataset, traditional CNN-based feature aggregation methods such as SCBANet and MFNet achieved mIoUs of 72.47% and 83.38%, respectively. While traditional CNN models have achieved some success in semantic understanding tasks, their reliance on local features and lack of global context modeling capabilities lead to a subpar performance in handling complex backgrounds and long-range dependencies. In contrast, CADANet, through the AFA and CAASA modules, effectively integrates global and local information, significantly improving semantic understanding accuracy. Transformer-based methods such as UNetFormer and FTUnetFormer capture global dependencies through self-attention mechanisms and excel in handling heterogeneous attribute differences. However, these models often overlook local details and background noise, resulting in insufficient detail understanding. CADANet addresses this issue through the CAASA module, integrating global background information and relative positional priors to enhance performance in complex background scenarios. Fig. 6(b) demonstrates CADANet's category-specific activation maps on the ISPRS Potsdam dataset, further validating its precise capture capability for various land-cover categories. Fig. 8 displays the visualization results of different models on the ISPRS Potsdam dataset, once again demonstrating CADANet's superior performance in complex scenes.

The experimental results on the WHDLD dataset demonstrate that CADANet excels in segmentation tasks across multiple categories. Compared to other state-of-the-art methods, CADANet achieves superior performance in categories,

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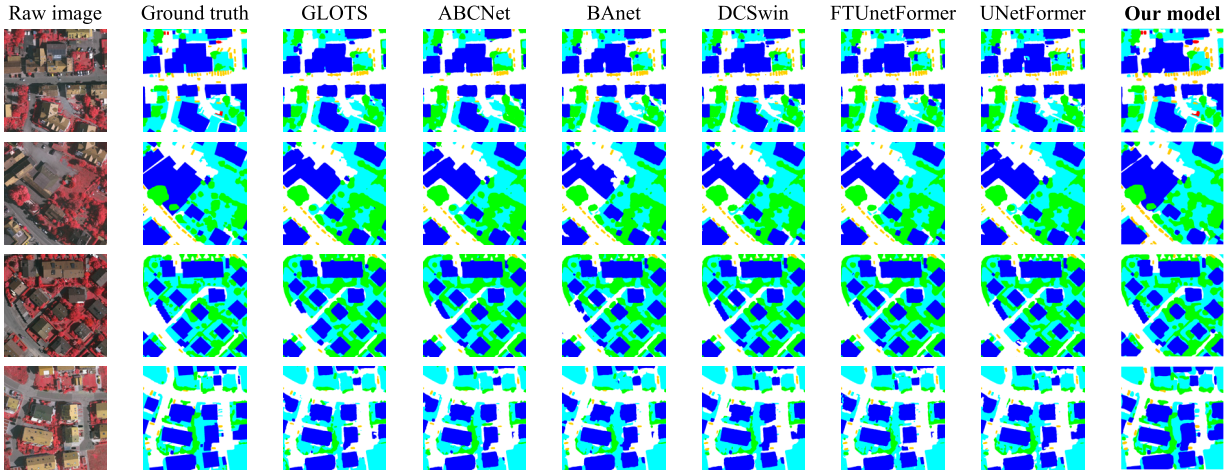


Fig. 7. Visual comparison of the experimental results on the Vaihingen dataset.

TABLE V

COMPARISON EXPERIMENT RESULTS (%) WITH STATE-OF-THE-ART METHODS ON THE ISPRS POTSDAM DATASET. THE BOLD MEANS THE BEST PERFORMANCE. UNDERLINE REPRESENTS SECOND BEST PERFORMANCE

Method	IoU (%)					mIoU (%)	mAcc(%)	mF1(%)
	ImSurf	Building	LowVeg	Tree	Car			
MGNet[50]	79.02	88.05	66.81	69.55	79.84	70.79	82.54	82.20
SCBANet[28]	-	-	-	-	-	72.47	82.37	82.88
BIBED-Seg[51]	87.40	90.40	75.30	79.20	67.30	76.00	90.20	-
PACSCNet[43]	82.58	90.41	73.82	74.12	91.15	76.03	85.21	-
Swin-SASPP[12]	84.52	92.4	75.07	76.88	82.66	75.10	82.79	-
GLOTS[18]	85.20	92.19	75.23	76.97	84.37	76.02	83.66	-
MFNet[26]	87.20	91.40	84.10	81.60	78.61	83.38	90.21	90.86
CMTFNet[16]	-	-	-	-	-	83.57	89.89	90.93
CMGFNet[41]	-	-	-	-	-	84.53	-	91.40
RSSFormer[32]	-	-	-	-	-	-	91.25	89.81
BAnet[45]	87.86	92.58	77.23	79.14	92.97	85.96	90.96	92.31
UNetFormer[42]	87.71	92.64	77.64	79.25	92.60	85.97	90.97	92.33
FTransUNet[44]	-	-	-	-	-	86.20	-	92.41
ABCNet[47]	87.57	92.27	78.11	78.89	91.34	85.82	91.03	92.27
EIGNet[19]	-	-	-	-	-	86.53	91.08	92.64
DCSwin[17]	89.25	93.83	78.86	81.01	92.69	87.13	91.81	<u>93.00</u>
FTUnetFormer[33]	89.21	**94.35**	<u>78.89</u>	<u>81.25</u>	<u>93.23</u>	<u>87.39</u>	<u>91.86</u>	93.15
CADANet(our)	**89.33**	<u>94.26</u>	**79.01**	**81.87**	**93.46**	**87.59**	**92.04**	**93.20**

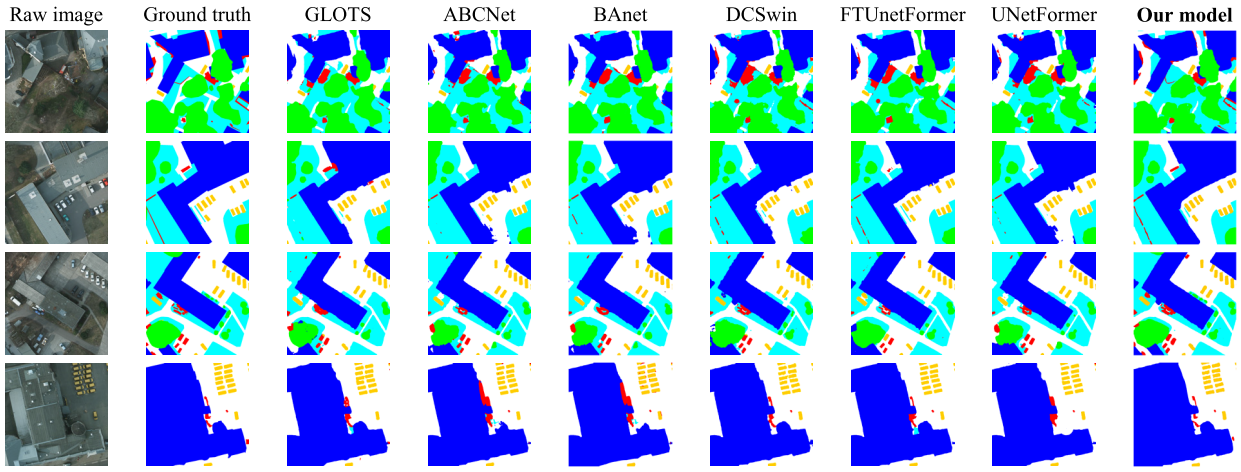


Fig. 8. Visual comparison of the experimental results on the Potsdam dataset.

such as buildings, roads, pavements, vegetation, bare soil, and water bodies. Specifically, as shown in Table VI, CADANet achieves IoUs of 57.97%, 62.86%, and 45.31% for buildings, roads, and pavements, respectively, and IoUs of 83.55%, 44.02%, and 94.93% for vegetation, bare soil, and water bodies, respectively. Overall, CADANet attains mIoU,

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TABLE VI

COMPARISON EXPERIMENT RESULTS (%) WITH STATE-OF-THE-ART METHODS ON THE WHDLD DATASET. THE BOLD MEANS THE BEST PERFORMANCE. UNDERLINE REPRESENTS SECOND BEST PERFORMANCE

Method	IoU (%)						mIoU (%)	mAcc(%)	mF1(%)
	Building	Road	Pavement	Vegetation	Bare soil	Water			
MFNet[53]	22.21	17.52	0	60.99	21.96	76.01	33.11	49.48	52.86
ABCNet[47]	35.30	42.52	29.79	77.13	30.80	87.52	43.11	80.97	58.39
BANet[45]	55.22	53.52	39.30	83.35	39.59	93.26	54.19	86.55	68.99
UNetFormer[42]	56.78	59.15	38.85	83.40	36.08	93.85	54.85	86.95	69.34
DCSwin[17]	55.84	56.52	37.27	83.06	42.18	93.22	54.97	86.50	69.65
FTUnetFormer[33]	58.94	59.47	40.82	**84.51**	43.51	94.29	57.45	**87.72**	71.79
CTMU-Net[52]	59.00	63.00	45.00	81.00	41.00	**95.00**	-	76.00	76.78
BIBED-Seg[51]	56.50	57.60	40.40	79.30	42.10	91.90	61.30	81.50	-
MGNet[50]	-	-	-	-	-	-	64.22	77.86	**78.19**
P-Swin[49]	**59.53**	**63.29**	45.16	81.11	43.99	94.58	64.61	84.86	-
CADANet(our)	57.97	62.86	**45.31**	83.55	**44.02**	94.93	**64.77**	85.24	77.55

mAcc, and mF1 scores of 64.77%, 85.24%, and 77.55%, respectively.

Traditional neural networks such as MFNet and ABCNet struggle with long-distance dependencies and complex background information. These methods primarily focus on extracting local features while neglecting the integration of global contextual information. For example, MFNet achieves IoUs of only 22.21% and 17.52% for buildings and roads, respectively, which are significantly lower than those achieved by CADANet.

Hybrid methods combining convolutional and transformer models, such as UNetFormer, DCSwin, and FTUnetFormer, partially improve the capture of long-distance dependencies by introducing the self-attention mechanism of transformer models. However, these methods still fall short in handling homogeneous and heterogeneous attribute differences. For instance, UNetFormer achieves IoUs of 59.15% and 38.85% for roads and pavements, respectively, while DCSwin achieves IoUs of 55.84% and 56.52% for buildings and roads, respectively, which are still significantly lower than CADANet's performance.

CADANet introduces the AFA and CAASA modules, effectively combining global background information and relative positional priors to capture long-distance dependencies between pixels. This significantly reduces the impact of homogeneous and heterogeneous attribute differences. For example, CADANet achieves an IoU of 94.93% for water bodies and 44.02% for bare soil, both surpassing other methods.

Although P-Swin also achieves commendable results, particularly in categories, such as vegetation (with an IoU of 81.11%) and water bodies (94.58%), the overall performance metrics of CADANet—mIoU of 64.77%, mAcc of 85.24%, and mF1 of 77.55%—demonstrate its broader generalization across multiple categories. While the improvements over P-Swin in some categories may appear less significant, CADANet's design is not solely focused on specific categories but rather on a comprehensive strategy that integrates both local and global contextual information. This allows CADANet to handle a wider range of segmentation tasks, particularly in addressing homogeneous and heterogeneous attribute differences, where traditional methods struggle.

Moreover, CADANet's design incorporates the AFA and CAASA modules, which enhance the model's ability to capture not only long-range dependencies but also local context, providing a more balanced approach to feature extraction. The generalization capability of CADANet across complex scenes is thus strengthened, allowing it to perform consistently well in various categories, including challenging areas, such as bare soil and pavements.

As illustrated in Fig. 9, the visual results of CADANet on the WHDLD dataset further validate its superior performance. When comparing CADANet with P-Swin, although P-Swin produces acceptable segmentation results in categories, such as water bodies and vegetation, CADANet consistently achieves more precise segmentation boundaries. Specifically, CADANet performs better in differentiating complex boundaries in built-up areas (as seen in the first row of Fig. 9) and vegetation-rich areas (second and third rows). In comparison, P-Swin's segmentation of buildings and bare soil appears more fragmented, with less accurate delineation of boundaries, resulting in a coarser segmentation output. CADANet also demonstrates a superior ability to handle fine-grained details and complex spatial relationships between different land-cover types, particularly in urban and semi-urban environments. For example, in the second row of Fig. 9, CADANet accurately distinguishes between roads, vegetation, and bare soil, whereas P-Swin displays more boundary blending and classification errors. Furthermore, in the third row, CADANet excels in maintaining the integrity of long, narrow structures such as roads, where P-Swin's segmentation output tends to blur these features. This advantage is largely due to CADANet's integration of global and local contextual information and the use of the self-attention mechanism, which allows the model to capture subtle details and long-distance dependencies more effectively. Ultimately, this enables CADANet to produce sharper, more coherent segmentations, even in complex scenes, enhancing the overall performance of semantic understanding tasks in remote sensing images.

We compared our proposed CADANet with several existing models based on the number of parameters (Params) and floating point operations per second (GFLOPs). As shown in Table VII, CADANet demonstrates significant advantages in both metrics. Specifically, CADANet has only 4.69 M

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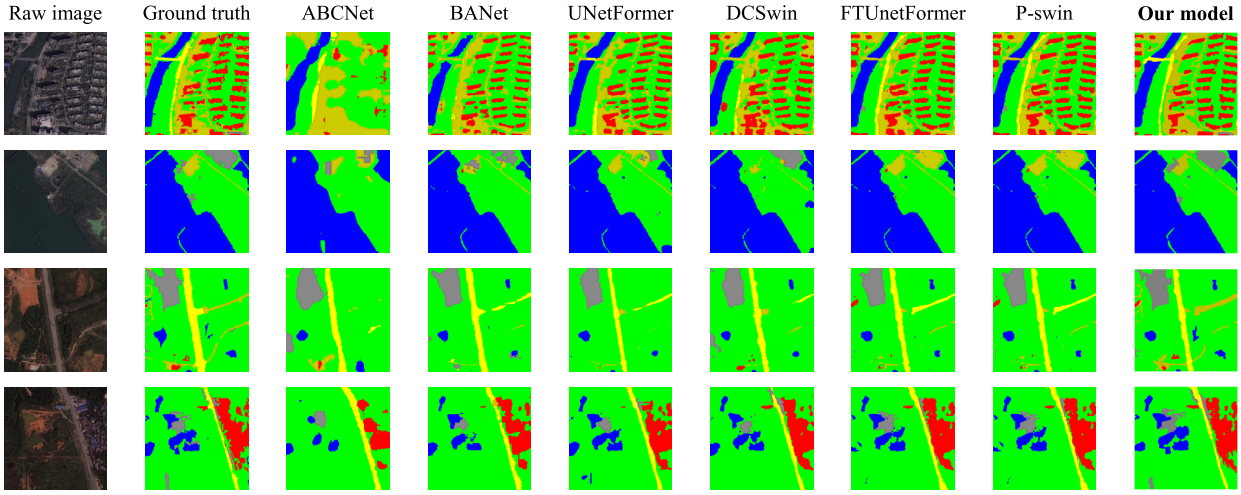


Fig. 9. Visual comparison of the experimental results on the WHDLD dataset.

TABLE VII
MODELS' NUMBER OF PARAMS AND GFLOPS

Method	Params (M)	GFLOPs
UNetFormer [42]	11.68	46.83
ABCNet44]	13.96	62.34
CMGFNet[41]	30.07	32.85
EIGNet[19]	34.48	318.8
MGNet[50]	45.79	**14.62**
BIBED-Seg[51]	78.69	448.59
FTUnetFormer[33]	95.98	497.98
DCSwin[17]	118.93	490.96
EMIDI[20]	122.27	235.95
P-Swin[49]	224.07	59.83
CADANet(our)	**4.69**	45.31

parameters, which is considerably lower than other models. For instance, UNetFormer [42] has 11.68 M parameters, ABCNet [47] has 13.96 M, and P-Swin [49], the model with the highest number of parameters, reaches 224.07 M. Compared to these models, CADANet reduces the number of parameters by approximately 60% to 97%, highlighting its notable lightweight design.

In terms of GFLOPs, CADANet's GFLOPs are 45.31 G, which is significantly lower than most other models. For example, DCSwin [17] has 490.96 G GFLOPs, FTUnetFormer [33] has 497.98 G, and BIBED-Seg [51] has as much as 448.59 G. This indicates that CADANet is more computationally efficient, reducing the computation load by about 90%–91% compared to these models. This low computational cost gives CADANet a clear advantage when processing high-resolution remote sensing images.

CADANet's unique strengths lie in its efficient module design. First, the AFA module combines global and local contextual information to effectively capture fine-grained features and contextual dependencies of similar attributes, thereby mitigating the impact of homogeneous attribute differences. Second, the CAASA module incorporates pixel-level global background information and relative positional priors, utilizing a self-attention mechanism to capture long-distance dependencies, thus addressing heterogeneous attribute differences. These innovative module designs not only enhance understanding performance but also significantly reduce the number of parameters and computational load.

TABLE VIII
ABLATION STUDY RESULTS (%) WITH DIFFERENT MODULES OF PARAMS AND GFLOPS

AFA	CAASA	CA	Params (M)	GFLOPs
×	✓	×	4.69	44.50
✓	×	✓	2.88	41.23
✓	✓	×	4.69	45.31

As shown in Table VIII, CADANet maintains a parameter count of 4.69 M when the AFA module is included, with only a 1.82% increase in GFLOPs compared to models without the AFA module. This outcome demonstrates that the AFA module enhances the model's ability to handle homogeneous attribute variation without adding to the model's parameter count, contrasting favorably with other information aggregation methods, such as GCNs. The parameter efficiency of this design means that the AFA module can improve consistency across similar land-cover types without significantly increasing computational load, enabling the model to deliver more precise results when processing features within similar categories.

Furthermore, replacing the CA module with the CAASA module, as observed in Tables I and II, adds just 1.81 M parameters while yielding substantial gains in mIoU, mAcc, and mF1. This indicates that while this configuration slightly increases computational complexity, it provides the model with a robust integration of both homogeneous and heterogeneous contextual information. The resulting CADANet excels at capturing both global and local features within complex remote sensing scenes, significantly improving the model's performance in understanding intricate land-cover characteristics.

V. CONCLUSION

This article introduces a novel CADANet designed to address the challenges of homogeneous and heterogeneous attribute differences in remote sensing image understanding. By incorporating the AFA module and the CAASA module, CADANet effectively combines global and local contextual information to capture fine-grained features and

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long-distance dependencies, thereby enhancing the model's semantic understanding performance. Extensive experiments on the ISPRS Vaihingen, ISPRS Potsdam, and WHDLD datasets demonstrate that CADANet significantly outperforms existing state-of-the-art methods in terms of accuracy and robustness. This innovative architecture not only showcases its powerful capabilities in high-resolution remote sensing image processing but also provides new avenues for future research. In upcoming studies, we aim to explore additional feature extraction and context modeling methods to further improve the performance of remote sensing image semantic segmentation. We also plan to investigate the application of semi-supervised and weakly supervised learning methods in scenarios with limited labeled data to enhance the model's generalization ability.

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